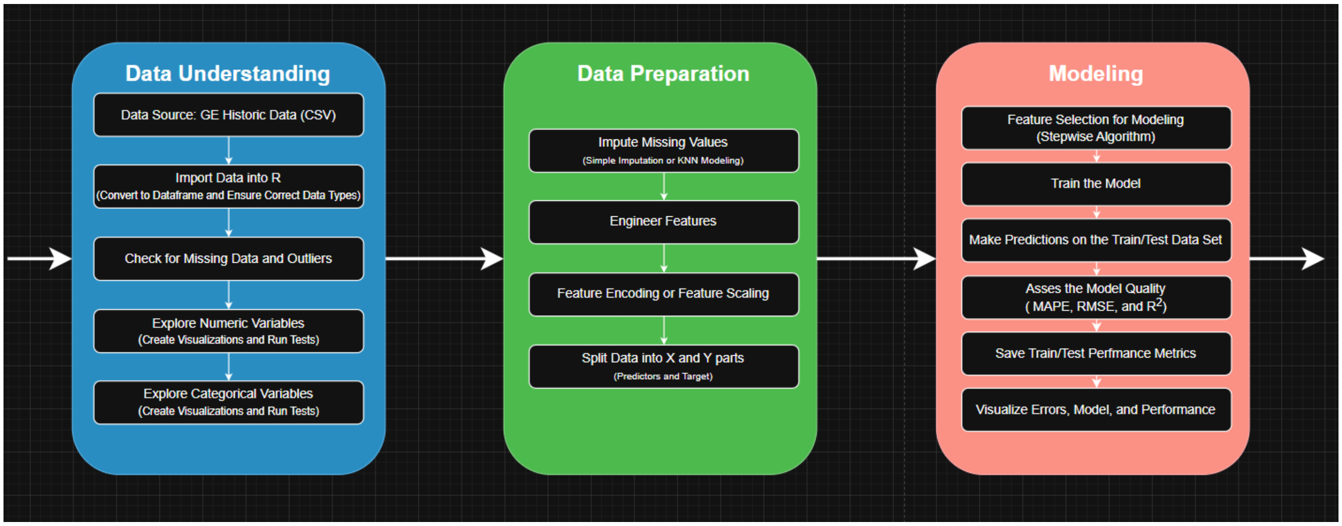

Above is the final version of my flowchart for the data modeling process that took place throughout this project. It is split into 3 sections that follow the CRISP-DM framework for managing a data science project. This flowchart shows the Data Understanding, Data Preparation, and Data Modeling phases, while acknowledging the Business Understanding that comes before it, and the Evaluation and Deployment that comes after it.

Data Understanding is exactly as it sounds; it involves the collection, exploration and verification of the data's quality. The steps I have that aligned with this phase were sourcing the CSV file from GE and importing it into R for exploration and quality check. GE also provided a data dictionary which was incredibly useful in the understanding of what the variables in this set meant. I used quartiles and summaries to determine what dating was missing and what was considered an outlier. I used visualization to explore the data, looking at correlations, distributions, and plots against the target variable. The knowledge gained from these tests and charts helped informed the direction I would take in the data preparation process. Not specifically mentioned in the flowchart but a step that was implemented here was the removal of all data that did not correlate to the target variable below a threshold.

The second phase of this chart outlines the process for my data preparation. The outcome of these processes sets up the data for all of the modeling. As such, the data cannot contain the errors, outliers. To start, I impute all of the missing values with the median value for the column. Median imputation is chosen over average imputations due to its robustness against outliers. It is a safer option, without losing any accuracy. Feature engineering would have come next, however, I did not end up creating any new features for this model. Many of the ratios I might have made already existed in the data. None of the features needed encoding as they were all numeric, but that step would have occurred next. The features were scaled logarithmically to reduce the of the large peaks that were visible in the data understanding charts. This logarithmic scaling helped a line fit to the data as well as seen in some simple A/B testing. The last step that sets up the data for modeling is to split it into X and Y parts. The X dataset contains all of the predictors, or dependent variables, while the Y part contains the target variable, or independent variable (loan amount).

The modeling phase began with the implementation of a stepwise feature selection algorithm. This step helps narrow down the correlated features to ones that actually contributed to the model's explanation of the data without making it needlessly complicated. Once the feature set was complete, the model was trained on the data using a linear regression. Predictions were made on both the dataset used for training, as well as the dataset given by GE to be used for validation. These predictions were quantified using various metrics like root mean squared error, which identified how far off the model was on average in dollar amounts, mean average percentage error which measured how far off the predicted loan value was in terms of percentage, and R2 which explains how well the model can explain the data. The redisuals were visualized in distribution charts and Q-Q plots to better understand where the model fell short.

And finally, saving all of these metrics and visualization will be a crucial part in evaluating further model iterations. They will be used as a baseline to compare the models accuracy and explanation of the dataset.

All of these steps combine to give me a model that has one of the best chances at succeeding. These phases then lead into model evaluation which will consider whether the model performs well enough to move towards deployment, or if it needs to be sent back to be reworked. Part of the evaluation phase too is consider whether or not the model fulfills the business objectives as well and meets all the requirements laid down at the beginning of the project .